Skills Summary

Arithmetic Sequences as Linear Functions

Use this reference while completing your worksheet or when studying.

Three questions, three tools. For an arithmetic sequence with first term a_1 and common difference d:

- *What is term n?* $a_n = a_1 + (n-1)d$
- *Which term is this value?* solve $a_1 + (n-1)d = \text{value}$ for n
- *What is the total of n terms?* $S_n = \dfrac{n(a_1 + a_n)}{2}$

Read the question first: a term, a term *number*, and a total are three different answers.

1. The explicit formula for a term

$a_n = a_1 + (n-1)d$. The $n-1$ is the count of **steps**, not terms: reaching term 10 from term 1 takes nine steps. A decreasing sequence simply has a negative d — the formula never changes.

Example: Find the 10th term of the sequence with $a_1 = 6$ and $d = 4$.
Step 1. One specific term is wanted, so use the explicit formula and count steps rather than listing all ten terms.
Step 2. Nine steps separate term 1 from term 10, so $a_{10} = 6 + (10-1)(4) = 6 + 36$.

$$a_{10} = \mathbf{42}$$

Try it: Find the 8th term of the sequence with $a_1 = 11$ and $d = 5$.

check: **46**

2. Solving for which term a value is

When the value is known and the position is not, set the explicit formula equal to the value and solve for n.

n must be a **positive whole number**. A fractional n means the value is not a term of the sequence at all — that is a real answer, not a mistake.

Example: Which term of $4, 10, 16, \ldots$ equals 58?
Step 1. The value is known but its position is not, so this is an equation to solve rather than a formula to evaluate.
Step 2. $4 + (n-1)(6) = 58$ becomes $6n - 2 = 58$, so $6n = 60$.

$$n = \mathbf{10}$$

Try it: Which term of $9, 16, 23, \ldots$ equals 65?

check: **9**

3. Summing a finite arithmetic sequence

$S_n = \dfrac{n(a_1 + a_n)}{2}$ — the number of terms times the **average** of the first and last.
The sum formula needs the *last* term, so compute a_n first whenever the problem gives only a_1, d and n.

Example: Find the sum of the first 10 terms of $5, 8, 11, \ldots$
Step 1. A total is wanted, so the sum formula applies — but it needs the last term, which is not given, so find that first.
Step 2. $a_{10} = 5 + 9(3) = 32$, and then $S_{10} = \dfrac{10(5+32)}{2} = 5 \times 37$.

$$S_{10} = \mathbf{185}$$

Try it: Find the sum of the first 12 terms of $2, 6, 10, \ldots$

check: **288**

4. A sequence as a linear function

Expanding $a_1 + (n-1)d$ gives $f(n) = dn + (a_1 - d)$: a line with **slope** d, sampled only at $n = 1, 2, 3, \ldots$
So the common difference is the slope, and the constant is the "term zero" value one step *before* the first term — not a_1 itself. That is the whole difference between discrete and continuous linear change.

Example: The sequence $7, 12, 17, \ldots$ is $f(n) = 5n + 2$. Find $f(12)$.
Step 1. The sequence is already in function form, so the term number is just an input — no stepping needed.
Step 2. Substituting the term number gives $f(12) = 5(12) + 2 = 60 + 2$.

$$f(12) = \mathbf{62}$$

Try it: The sequence $5, 11, 17, \ldots$ is $f(n) = 6n - 1$. Find $f(9)$.

check: **53**

(!) Watch out: use $n-1$, not n — one extra common difference is the most common wrong answer on this topic. Keep the sign of d from the story. And never multiply the number of terms by the *last* term to get a sum: multiply by the average of the first and last.